FIELD PROJECT · METRIC SPACES

CAN DISTANCE DETECT A HOLE?

A visual investigation of Cauchy sequences, completeness, and metric structure.

Durga Deshmukh
TYBSc · S.P. College

THE QUESTION IS NOT “WHAT IS DISTANCE?”

WHAT DOES DISTANCE ALLOW US TO DISCOVER?

A metric tells us when two points are close.

But closeness has consequences:

- Which sequences appear to converge?
- Which limits actually exist inside the space?
- Can a space contain “missing points”?
- Does the geometry reveal hidden structure?

This project explores these questions through one phenomenon:
how changing the metric can change completeness without changing topology.

A METRIC SPACE IS MORE THAN A SET OF POINTS

A metric space is a pair (X, d), where

- X is a set of points
- d assigns a non-negative distance to every pair of points in X

$d: X \times X \to [0,\infty)$

Non-negativity
$d(x,y) \geq 0$

Identity:
$d(x,y)=0$ iff $x=y$

Symmetry
$d(x,y)=d(y,x)$

Triangle Inequality
$d(x,z) \leq d(x,y)+d(y,z)$

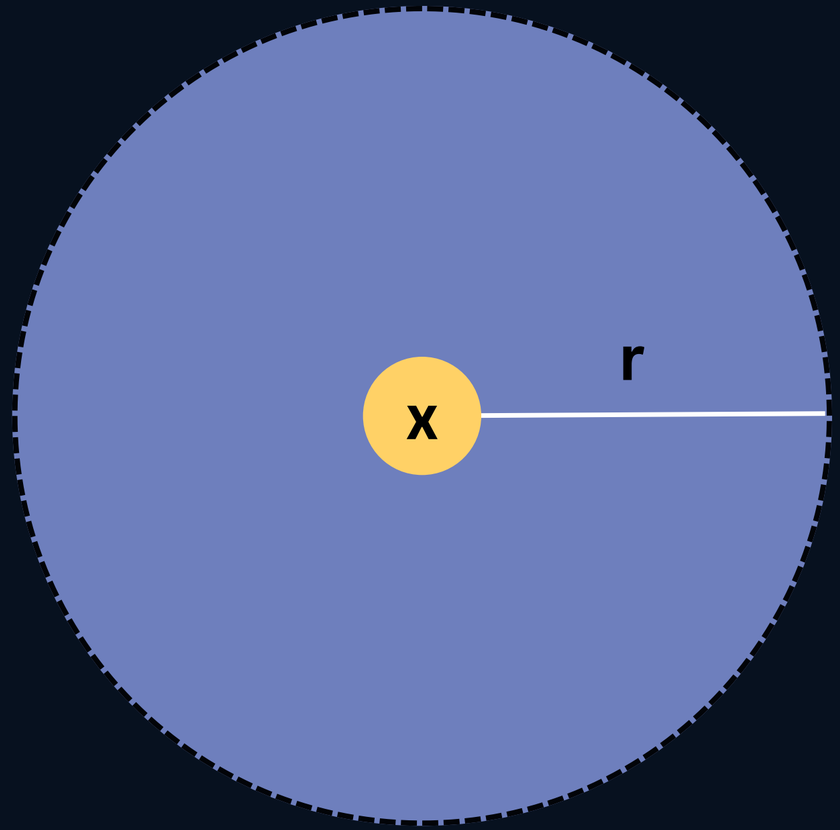
The metric is the rule that tells us what “close” means.

DISTANCE CREATES NEIGHBOURHOODS

For $x \in X$ and $r > 0$,

$$B(x,r) = \{ y \in X : d(x,y) < r \}$$

This is the open ball centred at x with radius r.



It answers one basic question:
Which points count as “close” to x?

All points inside satisfy $d(x,y) < r$

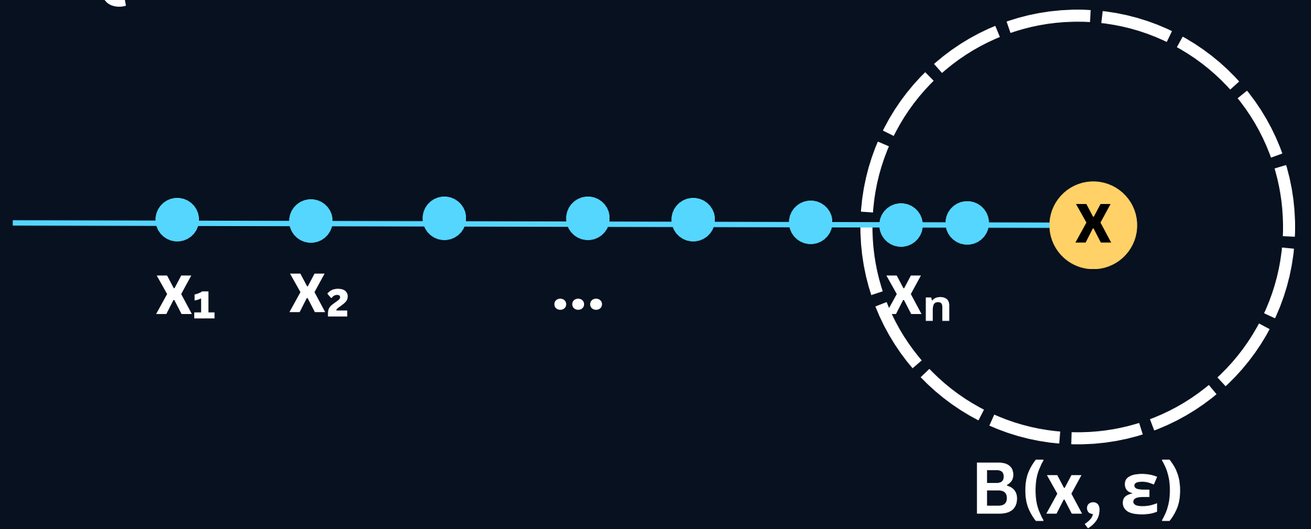
Open sets are built from open balls.
A set $U \subseteq X$ is open if every point $x \in U$
contains some open ball $B(x,r)$ entirely inside U.

WHEN DOES A SEQUENCE CONVERGE?

Convergent sequence

A sequence (x_n) converges to x in X if

$d(x_n, x) \to 0$ as $n \to \infty$



But what if we do not know the destination?

Cauchy Sequences

Terms eventually become close to one another.

$\forall \varepsilon > 0, \exists N$ such that $m,n \geq N \Rightarrow d(x_m,x_n) < \varepsilon$

Convergence asks: “Close to which point?”

Cauchy behaviour asks: “Close to each other?”

A CAUCHY SEQUENCE WITH NOWHERE TO GO

Take $X = \mathbb{Q}$ with $d(x,y)=|x-y|$.

Consider rational approximations to $\sqrt{2}$:

1, 1.4, 1.41, 1.414, 1.4142, ...

All terms belong to $\mathbb{Q}$.
The terms become arbitrarily close to one another.
Therefore, the sequence is Cauchy.

Inside $\mathbb{Q}$

$1 \to 1.4 \to 1.41 \to 1.414 \to 1.4142 \to \dots$

?

The limit is not in $\mathbb{Q}$

completion

Inside $\mathbb{R}$

$1 \to 1.4 \to 1.41 \to 1.414 \to 1.4142 \to \sqrt{2}$

The limit exists in $\mathbb{R}$

COMPLETENESS: WHEN EVERY APPROACH HAS A DESTINATION

(X,d) is complete ⇔ Every Cauchy sequence in X converges to a point of X.

$\mathbb{Q}$ under $d(x,y)=|x-y|$

↓

Cauchy sequence

↓

approaches $\sqrt{2}$

↓

$\sqrt{2} \notin \mathbb{Q}$

↓

Incomplete: a Cauchy sequence has no limit in the space.

$\mathbb{R}$ under $d(x,y)=|x-y|$

↓

Cauchy sequence

↓

approaches $\sqrt{2}$

↓

$\sqrt{2} \in \mathbb{R}$

↓

Complete: the limit belongs to the space.

Can we change the metric and fill the hole?

THE SAME SET: X=(0,1)

Usual metric: $d(x,y)=|x-y|$

The sequence $\{1/n\}$ approaches the missing endpoint 0.

INCOMPLETE

Transformed metric:

$\rho(x,y)=|h(x) - h(y)|$

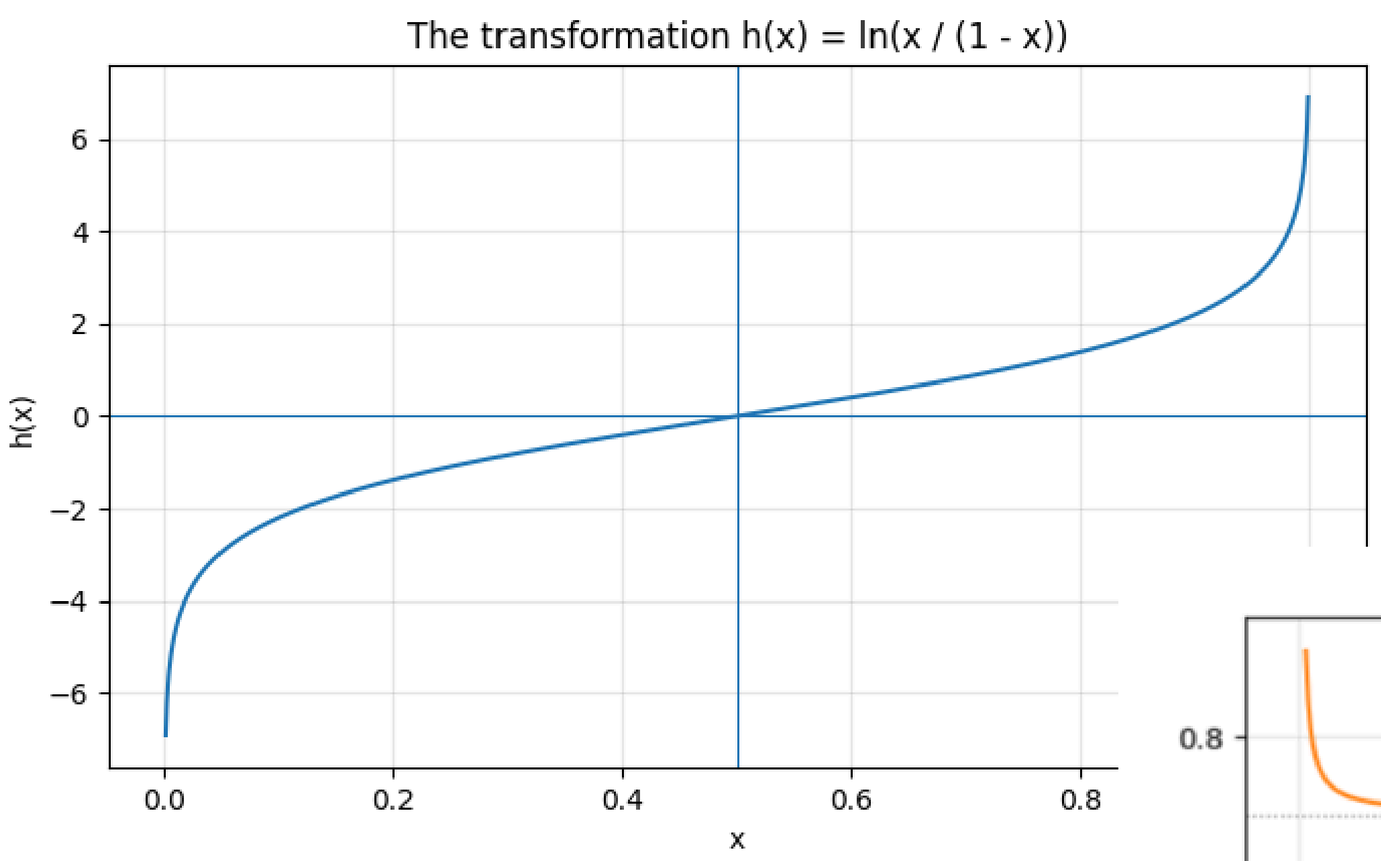
Where:

$h(x)=\ln(x / (1 - x))$

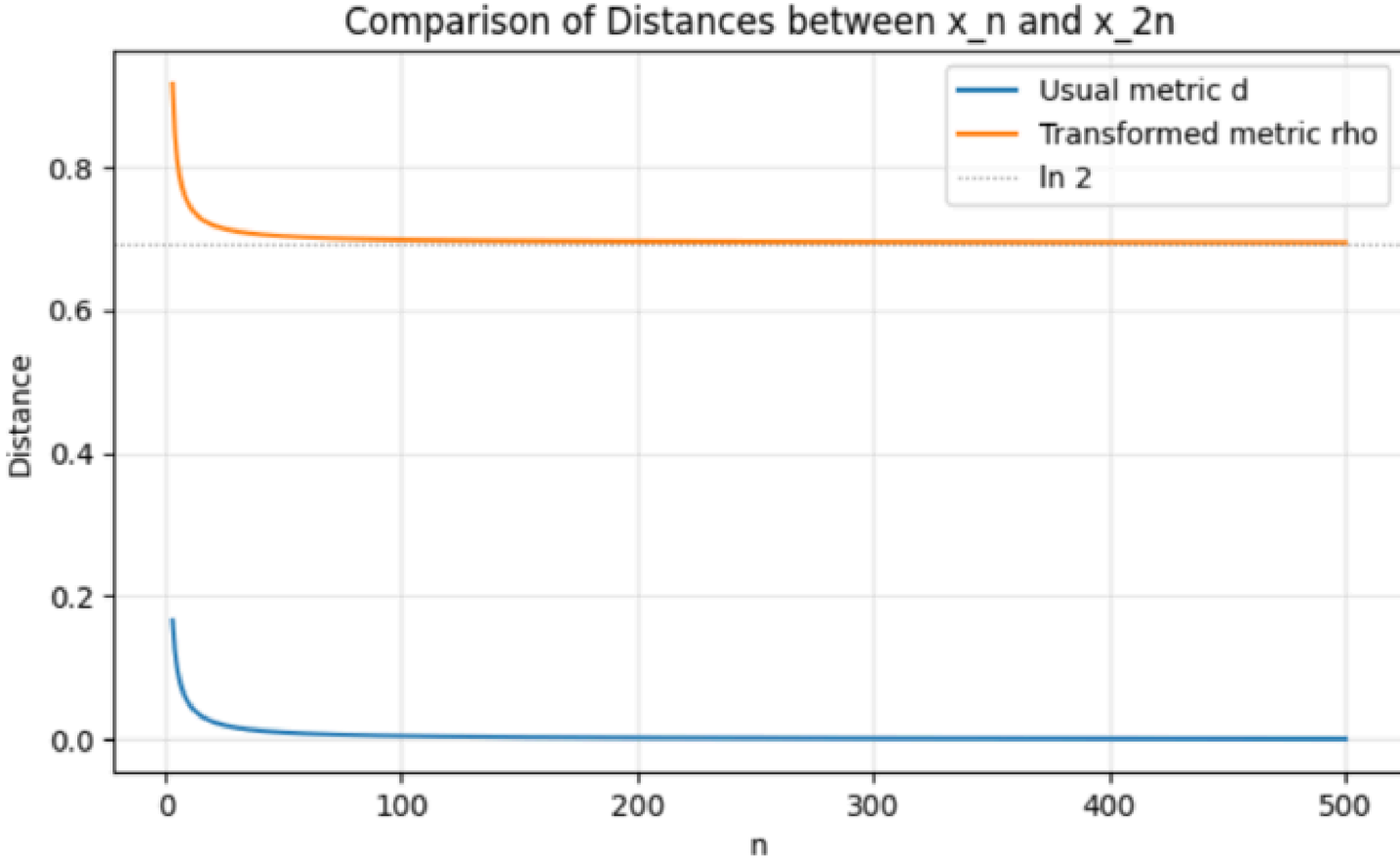
COMPLETE

Same set. Same topology. Different completeness.

The metric changes how distances are measured without changing the open sets.



$$d(x_n, x_{2n}) \to 0$$



$$\rho(x_n, x_{2n}) \to \ln 2$$

ρ IS ℝ IN DISGUISE

$h(x) = \ln(x/(1-x))$ is a bijection $(0,1) \to \mathbb{R}$, with $h'(x) = 1/(x(1-x)) > 0$ and inverse $g(t) = e^t/(1+e^t)$.

$\rho(x,y) = |h(x) - h(y)|$ is a metric: h is injective (identity axiom), and the triangle inequality is inherited from $\mathbb{R}$.

$h : ((0,1), \rho) \to (\mathbb{R}, |\cdot|)$ is a surjective isometry.

$((0,1), \rho)$ is $\mathbb{R}$ with its points renamed.

SAME OPEN SETS

Ball $B_\rho(x,r) = h^{-1}((h(x)-r, h(x)+r))$ is the preimage of an open interval under the continuous h, so it is d-open.



Conversely, if U is d-open, then $h(U) = g^{-1}(U)$ is open in $\mathbb{R}$. So for each $x \in U$ there is r with $B_\rho(x,r) \subseteq U$.

Consequence: the same convergent sequences, the same closed sets, the same continuous functions.

ρ IS COMPLETE, AND WHY

Proof: if (x_n) is ρ-Cauchy, then $(h(x_n))$ is Cauchy in $\mathbb{R}$, so $h(x_n) \to L$, so $x_n \to g(L) \in (0,1)$.

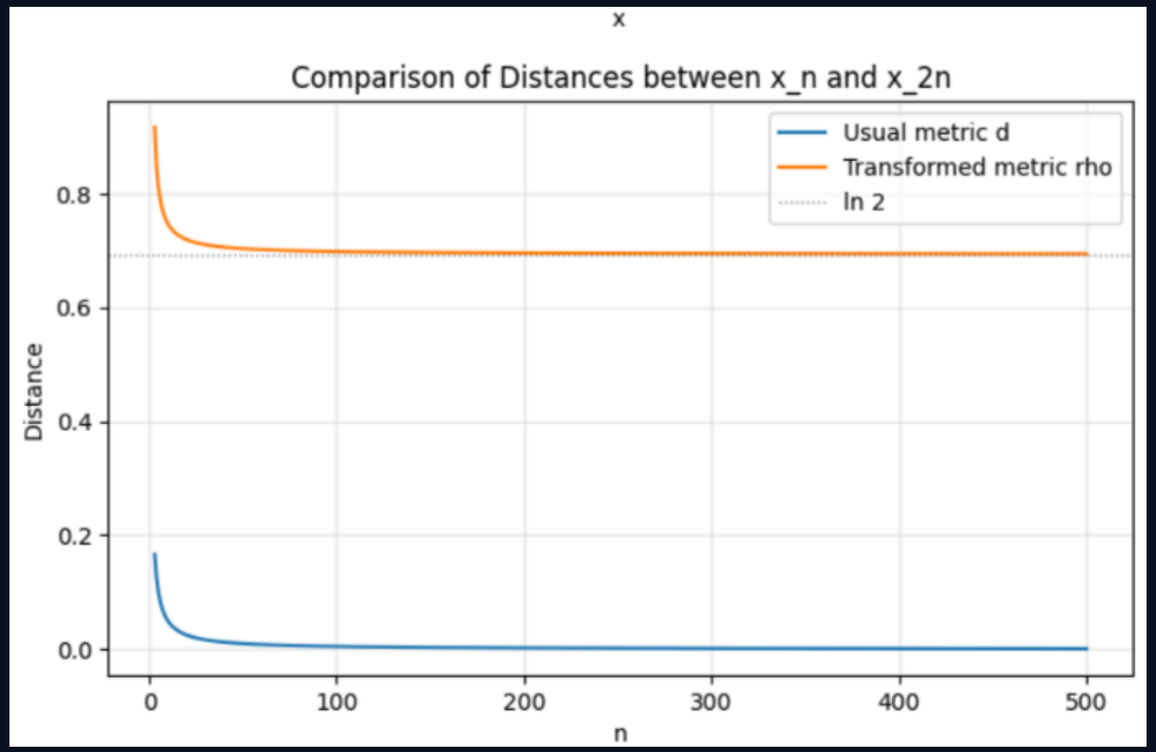
Usual d: the inclusion $(0,1) \subset \mathbb{R}$ has non-closed image, so d is incomplete.
ρ: the image is $\mathbb{R}$, which is closed, so ρ is complete.

Pullback criterion: For an injective $h : X \to Y$, the metric $d_h(x,y) = e(h(x), h(y))$ is complete $\Leftrightarrow$ $h(X)$ is a complete subspace. If Y is complete, this means $h(X)$ is closed in Y.

Completeness of a pulled-back metric is closedness of the image.

WHERE EXACTLY DO d AND ρ DIFFER?

Mean Value Theorem: $\rho(x,y) = h'(c)|x-y|$, and $h'(c) = 1/(c(1-c)) \geq 4$. So $\rho \geq 4d$.
Hence every ρ-Cauchy sequence is d-Cauchy.

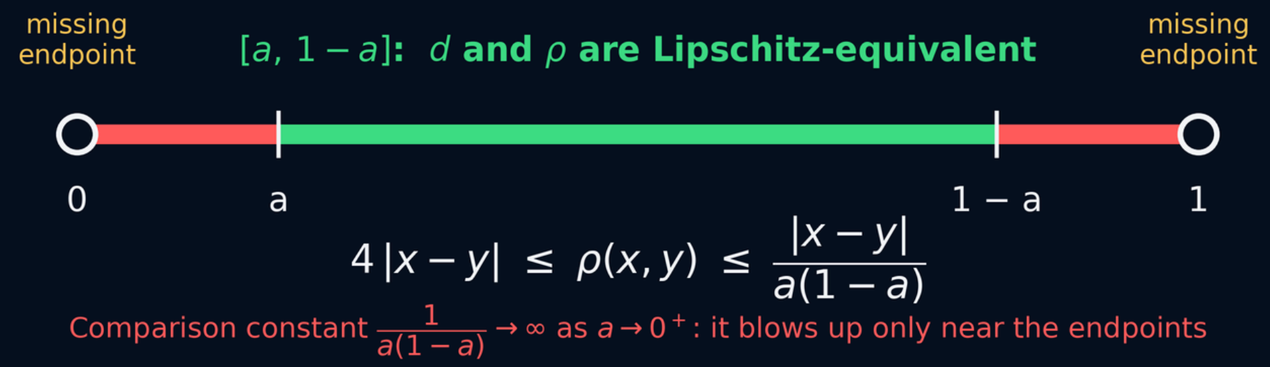


The converse fails: if (1/n) were ρ-Cauchy, then $\rho(x_n, x_{2n}) \to 0$. But $\rho(x_n, x_{2n}) = \ln((2n-1)/(n-1)) \to \ln 2$.

A HOMEOMORPHISM THAT IS NOT UNIFORMLY CONTINUOUS

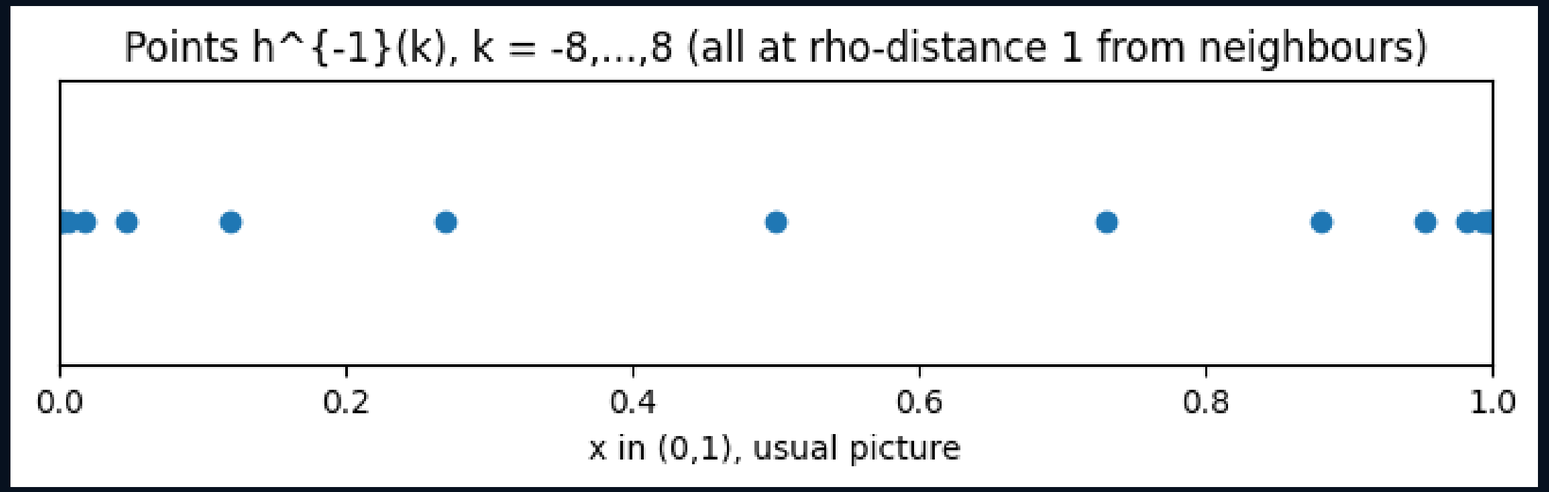
((0,1), d) → ((0,1), ρ) is continuous in both directions. It is not uniformly continuous, because uniformly continuous maps preserve Cauchy sequences and (1/n) breaks that.

On [a, 1−a]: 4|x−y| ≤ ρ(x,y) ≤ |x−y| / (a(1−a)).



The Lipschitz constant 1/(a(1−a)) blows up only at the missing endpoints, which is where the hole is.

THE RULER PICTURE



The points g(k), k = −8…8, are all at ρ-distance 1 from their neighbours.

ρ(½, 1/n) = ln(n−1) → ∞.

the endpoints are infinitely far away in ρ.

WHAT SURVIVES A CHANGE OF METRIC?

Property	d	ρ	Topological?
Open / closed sets, convergence, continuity	same	same	yes
Compact	no	no	yes
Cauchy sequences	1/n yes	1/n no	no
Complete	no	yes	no
Bounded, totally bounded	yes	no	no

Compact = Complete + totally bounded, and no metric inducing the usual topology can make (0,1) both.

A HIERARCHY OF EQUIVALENCES

Lipschitz ⇒ uniform ⇒ same Cauchy sequences ⇒ same completeness. Also uniform ⇒ topological. The pair (d, ρ) is only topological.

Set	Metrics	Same topology	Same Cauchy	Completeness
(0,1)	d vs ρ	yes	no	no → yes
$\mathbb{R}$	$\lvert\cdot\rvert$ vs $\lvert\arctan x - \arctan y\rvert$	yes	no	yes → no
$\mathbb{R}$	d vs min(d,1)	yes	yes	yes, yes
$\mathbb{R}^n$	d_1 vs d_2	yes	yes	yes, yes

$\sigma(x,y) = |\arctan x - \arctan y|$ has the same topology on $\mathbb{R}$, but $x_n = n$ is σ-Cauchy with no limit. So a metric change can also destroy completeness.

Completeness is a uniform property, not a topological one.

COMPLETIONS, AND WHY COMPLETENESS MATTERS

Completions: $(\mathbb{Q}, |\cdot|) \to \mathbb{R}$. $((0,1), d) \to [0,1]$, adding two points. $((0,1), \rho) \to$ itself, adding none.

Banach: $f(x) = x/2$ contracts $((0,1), d)$ with constant ½ and has no fixed point, since $0 \notin (0,1)$. No contradiction, because d is incomplete. Under ρ, Banach implies f cannot be a contraction.

The missing points depend on the metric.

CAN EVERY HOLE BE REMOVED?

U open in a complete (X,d), $\delta(x) = \mathrm{dist}(x, U^c)$. Then $\rho_U = d + |1/\delta(x) - 1/\delta(y)|$ is complete and has the same topology. This covers (0,1), $\mathbb{R}^n$ minus a point, and open discs.

Proof idea: if a ρ_U-Cauchy sequence converged to a point $z \notin U$, then $1/\delta(x_n) \to \infty$, which contradicts a Cauchy real sequence being bounded.

$\mathbb{Q}$ admits no complete compatible metric. If it did, $\mathbb{Q} = \bigcup\{q\}$ would be a countable union of closed nowhere dense sets, contradicting Baire.

The hole in (0,1) can be repaired by changing the metric. The holes in $\mathbb{Q}$ cannot be repaired by any metric.

CONCLUSION

- A metric determines Cauchy sequences, completeness and boundedness. The topology determines only the open sets.
- d and ρ induce the same topology but differ in completeness. The identity map is a homeomorphism that is not uniformly continuous, and the difference sits at the endpoints.
- Completeness is preserved by uniform equivalence, not topological equivalence, and a metric change can create or destroy it.
- Open subsets can be completed by re-measuring. $\mathbb{Q}$ cannot.
- Final insight: Completeness belongs to the metric, not to the topology.